

# Rewiring the market time clock in Spanish electricity markets: strategic behaviour and market power with higher granularity

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June 2026

**Abstract**



# 1 Introduction

## 1.1 Motivation and research question

## 1.2 Related literature

## 1.3 Mechanism summary and contribution

## 1.4 Roadmap

# 2 Theoretical model

## 2.1 Setup

## 2.2 Equilibrium under low granularity

## 2.3 Equilibrium under high granularity

## 2.4 Predictions: dominant vs fringe response in time-varying vs flat subperiods

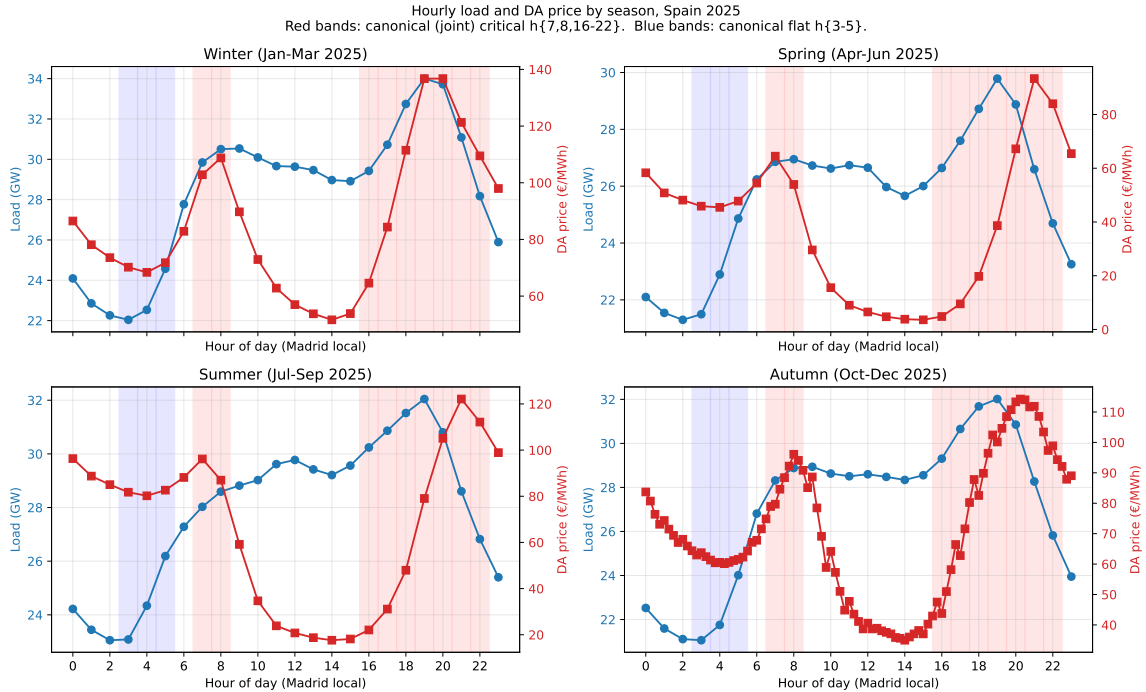
# 3 Institutional setting, data, and identification

## 3.1 Spanish wholesale electricity market: a sequential-market overview

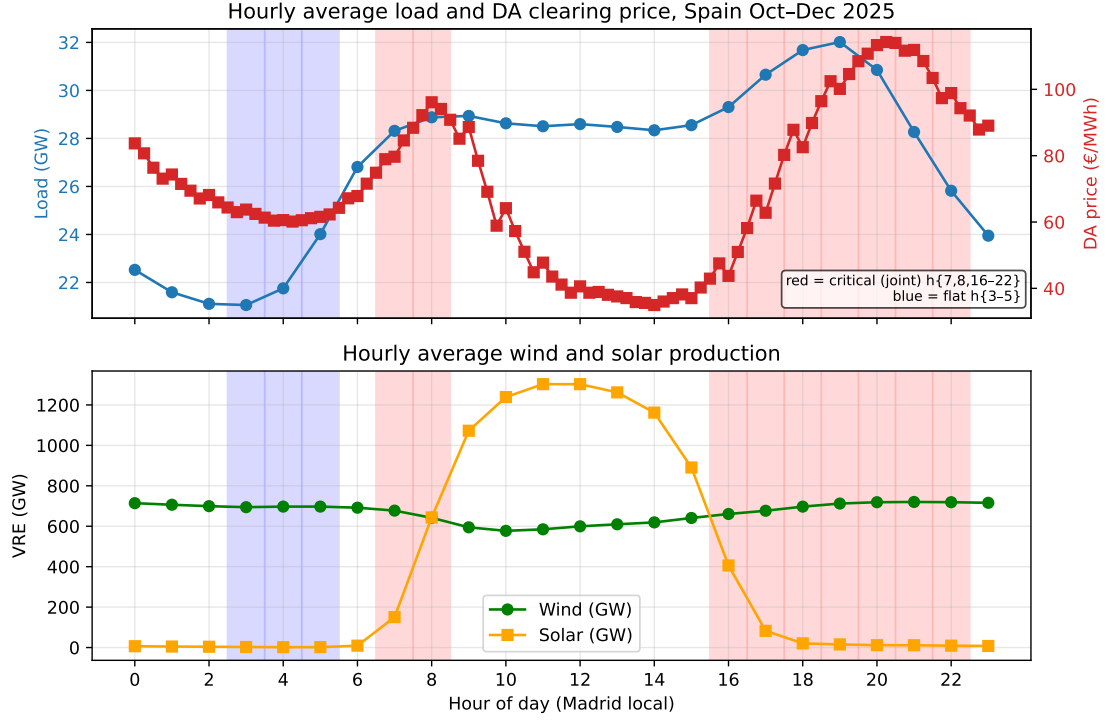
## 3.2 The MTU15 reform sequence

## 3.3 Data sources

## 3.4 Critical-vs-flat hours: empirical calibration



**Figure 1: Hourly load and DA price profile, by season, Spain 2025.** Four-panel decomposition by season. Red bands mark the canonical *critical hours*  $h \in \{7, 8, 16, 17, 18, 19, 20, 21, 22\}$  (the *joint* set: morning ramp  $\cup$  afternoon-evening peak). Blue bands mark the *flat hours*  $h \in \{3, 4, 5\}$ . The price peak  $h \in \{18-22\}$  is stable across all four seasons; spring and summer additionally show a morning ramp peak at  $h \in \{7-8\}$ . Flat hours are stable everywhere. The joint set is the natural cross-season-robust definition used in the headline DiD.



**Figure 2: Hourly profile of Spanish electricity, October–December 2025.** Top panel: average load (left axis, blue) and DA clearing price (right axis, red) by hour of day, Madrid local time. Bottom panel: average wind and solar production. Red bands mark the canonical *critical hours*  $h \in \{7, 8, 16, 17, 18, 19, 20, 21, 22\}$  (the *joint* set: morning ramp  $\cup$  afternoon-evening peak; peak load and peak DA price hours). Blue bands mark the *flat hours*  $h \in \{3, 4, 5\}$  (load and price minima, no solar). Critical hours combine high demand with low VRE supply; flat hours have low demand and no time variation in either direction.

### 3.5 Identification strategy and treatment-effect set

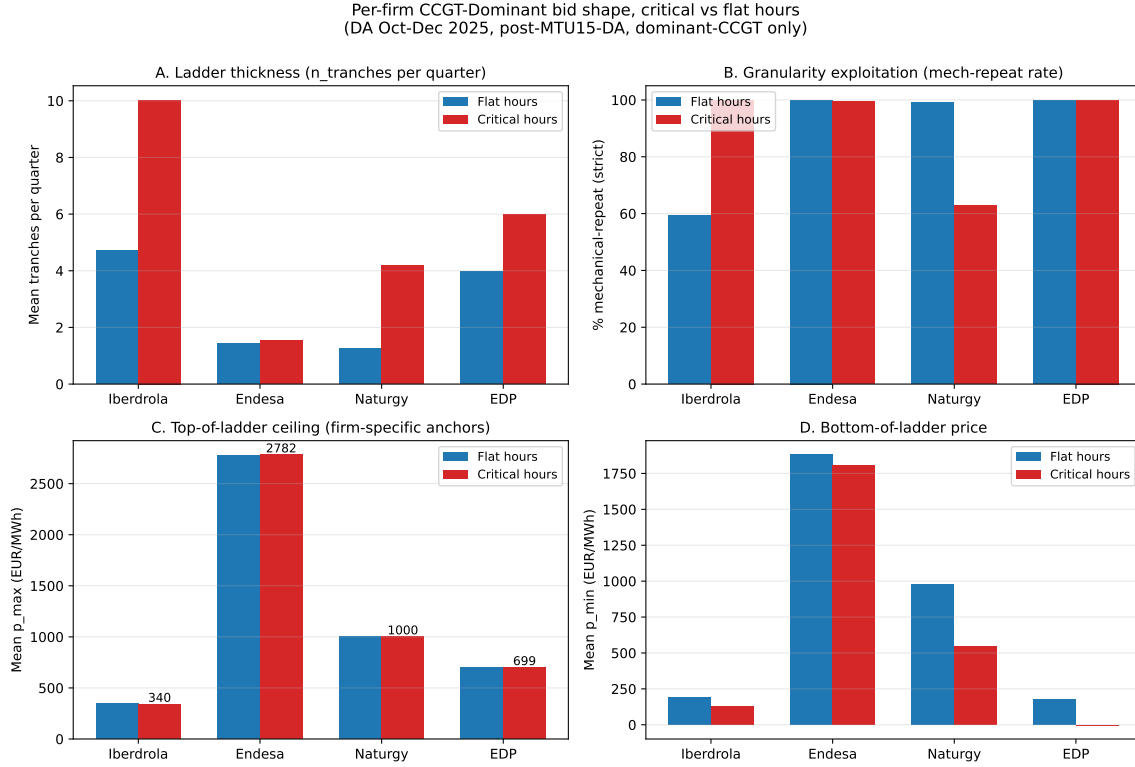
**Table 1: Pivotality rate by parent firm and hour-class, CCGT only, October–December 2025.** Pivotality is the unit-level price-setter rate: tranches straddle the DA clearing price (i.e.,  $n_{\text{below}} > 0$  and  $n_{\text{above}} > 0$ ), evaluated per (date, unit, period) and aggregated by parent firm  $\times$  hour-class. The treatment-effect set in the empirical specifications is the set of firms with non-trivial CCGT pivotality in critical hours (HC, EDP-PT, GN, GE, IB plus mid-fringe CCGT operators). The placebo set is firms with effectively zero pivotality (Repsol, Engie España, TotalEnergies, Moeve). Source: notebooks/memos/pivotality\_by\_firm\_critical\_hours.md.

| Firm (parent)           | flat $h\{3-5\}$ | critical $h\{18-22\}$ | $\Delta$ crit–flat |
|-------------------------|-----------------|-----------------------|--------------------|
| HC                      | 0.72            | 1.00                  | +0.28              |
| EDP-PT                  | 0.79            | 0.88                  | +0.09              |
| GN                      | 0.02            | 0.84                  | <b>+0.82</b>       |
| Other (mid-fringe CCGT) | 0.26            | 0.40                  | +0.14              |
| GE                      | 0.12            | 0.31                  | +0.19              |
| IB                      | 0.004           | 0.25                  | +0.25              |
| Engie España            | 0.00            | 0.005                 | 0.00               |
| Moeve                   | 0.005           | 0.005                 | 0.00               |
| Repsol                  | 0.00            | 0.00                  | 0.00               |
| TotalEnergies           | 0.00            | 0.00                  | 0.00               |

### 3.6 Parallel-trends discussion

## 4 Bidding-behavior evidence of the mechanism

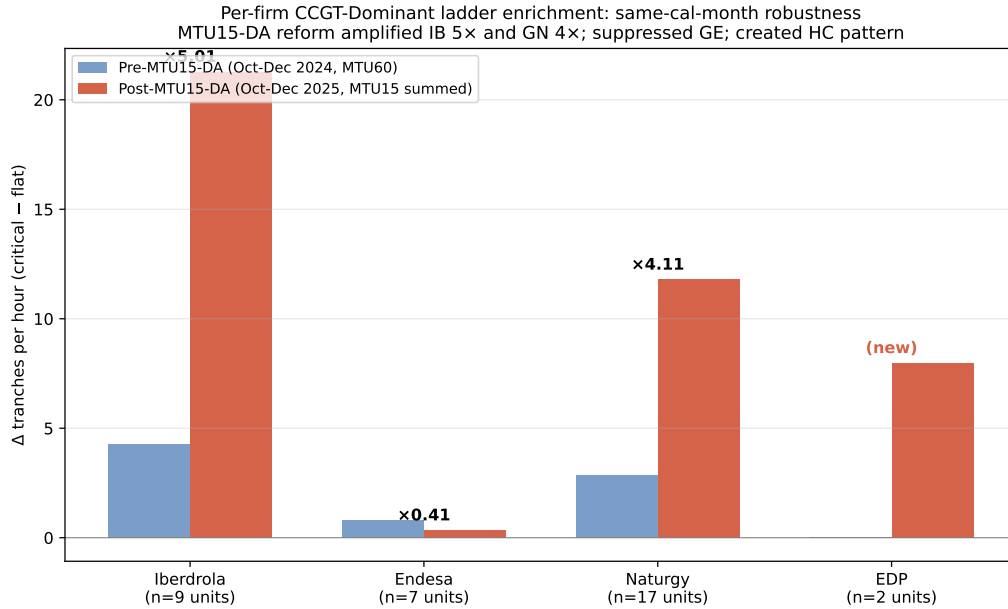
### 4.1 Per-firm bid-shape patterns at hour-of-day resolution



**Figure 3: Per-firm CCGT bid-shape patterns, October–December 2025 day-ahead market.** Four-panel decomposition of dominant CCGT bidding by firm (Iberdrola, Endesa, Naturgy, EDP-Spain). Panels show per-firm critical vs flat-hour means for tranche count, mechanical-repeat rate, top-of-ladder price, and bottom-of-ladder price. Reveals two distinct strategic types: Iberdrola (uniform enricher, richest ladders, no quarter variation) and Naturgy (granularity exploiter, varies bids quarter-by-quarter). Endesa and EDP-Spain show mid-strategic and inactive patterns respectively. Source: `scripts/analysis/firm/per_firm_hourly_ccgt_bidshape.py`.

## 4.2 Granularity exploitation: ladder, within-hour quarter variation, and their interaction

## 4.3 Pre- vs post-MTU15-DA structural changes



**Figure 4: Pre- vs post-MTU15-DA amplification of the critical-flat differential, by firm.** Same-calendar-month comparison (October–December 2024 vs October–December 2025) of CCGT bid-shape critical-flat differential by firm. Iberdrola amplifies 5.0×, Naturgy 4.1×, EDP-Spain creates a positive differential where there was none, while Endesa’s pattern is suppressed. Indicates asymmetric strategic response to the granularity reform across the dominant tier. Source: `scripts/analysis/bid/per_firm_pre_vs_post_mtu15da.py`.

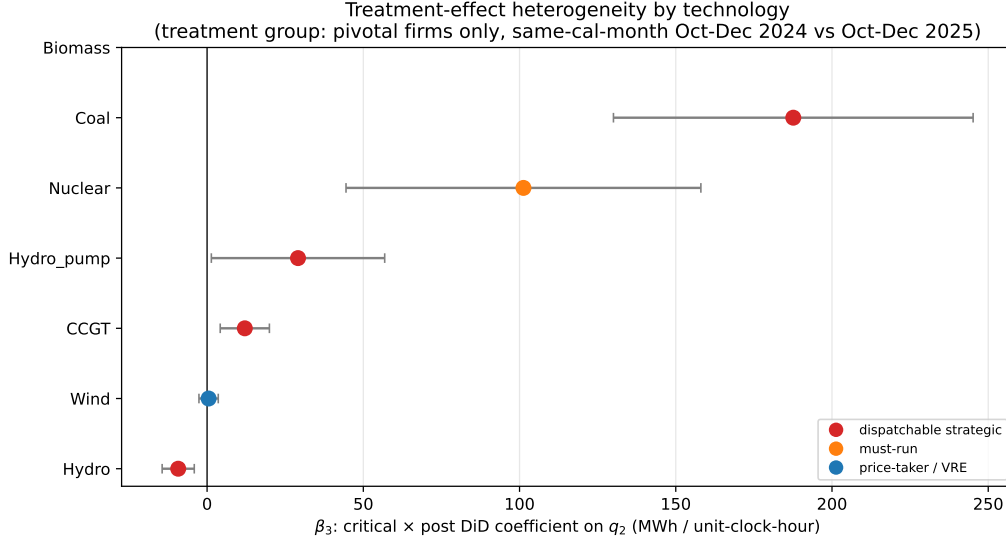
#### 4.4 Heterogeneity by firm and technology

### 5 Main empirical results

#### 5.1 Within-day DiD on $q_2$ : primary outcome

**Table 2: B1 – Within-day DiD on  $q_2$  (Ito–Reguant strategic IDA upward adjustment).** Outcome is  $q_2$  in MWh per (unit, clock-hour), defined as the sum of `pibci` across IDA sessions converted to MWh and aggregated to clock-hour resolution. Specification:  $q_{2,fdh} = \alpha + \beta_1 \text{crit}_h + \beta_2 \text{post}_d + \beta_3 (\text{crit}_h \times \text{post}_d) + \gamma_f + \delta_{\text{DOW}(d)} + \varepsilon_{fdh}$ . Firm and day-of-week fixed effects; cluster-robust standard errors by date. Sample: same-calendar-month windows, October–December 2024 and October–December 2025 (both windows post-MTU15-IDA reform of March 2025). Critical hours:  $h \in \{7, 8, 16, 17, 18, 19, 20, 21, 22\}$  (joint: morning ramp  $\cup$  evening peak; cross-season-robust). Flat hours:  $h \in \{3, 4, 5\}$ . Treatment group: firms with  $\geq 10\%$  pivotality in critical hours (HC, EDP-PT, GN, GE, IB). Placebo: firms with  $\sim 0\%$  pivotality (Repsol, Engie España, TotalEnergies, Moeve). Significance: \*  $p < 0.05$ , \*\*  $p < 0.01$ , \*\*\*  $p < 0.001$ ; headline interpretation relies on \*\*\* given the large effective sample size.

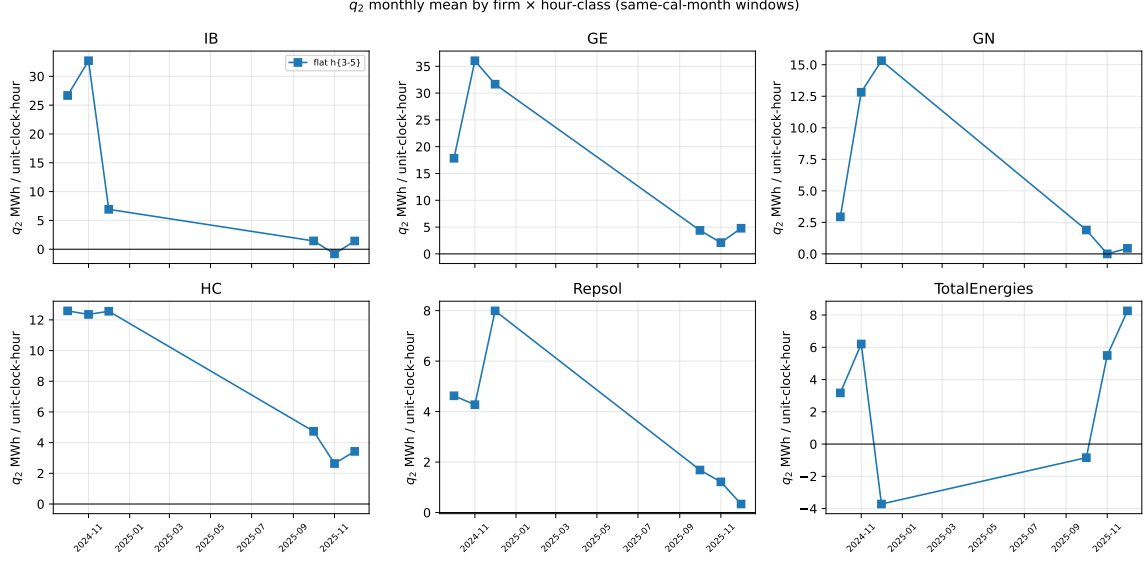
|                                                          | All firms<br>(1) | Treatment group<br>(2) | Placebo group<br>(3) |
|----------------------------------------------------------|------------------|------------------------|----------------------|
| $\beta_3$ : <code>crit</code> $\times$ <code>post</code> | +0.26*           | +7.56***               | −3.18***             |
| cluster-robust SE                                        | (0.12)           | (1.42)                 | (0.36)               |
| $p$                                                      | 0.0222           | 0.0000                 | 0.0000               |
| $\beta_1$ : <code>crit</code>                            | −0.28            | −7.89                  | +3.41                |
| $\beta_2$ : <code>post</code>                            | −0.25            | −15.08                 | −0.18                |
| $\bar{y}$ (pre, flat)                                    | 0.27             | 17.06                  | 0.34                 |
| $\bar{y}$ (pre, crit)                                    | −0.10            | 9.07                   | 3.79                 |
| $\bar{y}$ (post, flat)                                   | 0.02             | 1.81                   | 0.25                 |
| $\bar{y}$ (post, crit)                                   | −0.07            | 1.47                   | 0.55                 |
| $n$                                                      | 4,076,148        | 278,408                | 493,759              |
| clusters $G$ (dates)                                     | 184              | 184                    | 184                  |



**Figure 5: B1 – Treatment-effect heterogeneity by technology.** Coefficient plot of  $\beta_3$  (critical  $\times$  post DiD interaction) on  $q_2$ , stratified by `tech_group` within the treatment group. Error bars are 95% confidence intervals using cluster-robust standard errors by date. Color coding: red = dispatchable strategic technologies (CCGT, Hydro, Hydro\_pump, Coal); orange = must-run (Nuclear); blue = price-taker / VRE (Wind). The dispatchable–non-dispatchable split is the key validation of the within-day-DiD design: granularity has economic content for technologies that can strategically reshape supply, not for must-run or price-taker technologies. Source: Table 3.

**Table 3: B1 –  $\beta_3$  stratified by technology, treatment vs placebo group.** Same specification as Table 2, restricted by technology and group. The “Predicted” column shows the model’s directional prediction: + for dispatchable strategic technologies (CCGT, Coal, Hydro, Hydro\_pump), 0 for must-run (Nuclear) and price-taker (Wind, Biomass). The treatment-group dispatchable techs all show large positive  $\beta_3$ ; Wind shows the predicted null. The placebo group should show  $\beta_3 \approx 0$  across all technologies. Significance markers as in Table 2.

| Technology | Treatment group |                  | Placebo group |                  | Predicted |
|------------|-----------------|------------------|---------------|------------------|-----------|
|            | $\beta_3$       | SE / $n$         | $\beta_3$     | SE / $n$         |           |
| CCGT       | +12.08**        | (4.01) / 53,149  | +5.83         | (15.94) / 11,958 | +         |
| Hydro      | −9.24***        | (2.62) / 34,241  | +5.68         | (5.40) / 4,512   | +         |
| Hydro_pump | +29.10*         | (14.15) / 6,205  | —             | —                | +         |
| Coal       | +187.61***      | (29.37) / 639    | —             | —                | +         |
| Nuclear    | +101.25***      | (28.97) / 19,968 | —             | —                | 0         |
| Wind       | +0.49           | (1.58) / 72,122  | +0.76***      | (0.20) / 266,577 | 0         |
| Biomass    | —               | —                | —             | —                | 0         |



**Figure 6:  $q_2$  monthly mean by firm and hour-class, same-calendar-month windows.** Six-panel monthly trajectory of  $q_2$  (in MWh per unit-clock-hour) for the four treatment-group firms (Iberdrola, Endesa, Naturgy, EDP-Spain) and two placebo firms (Repsol, TotalEnergies). Red = critical hours  $h\{18-22\}$ ; blue = flat hours  $h\{3-5\}$ . The two same-calendar-month windows (October–December 2024 and October–December 2025) bracket the MTU15-DA reform of October 1, 2025. Both windows are post-MTU15-IDA so the IDA reform of March 2025 is held constant. Source: `scripts/analysis/firm/thesis_figures.py`.

**Table 4: Per-firm  $\beta_3$  on  $q_2$  (B1) and DA cleared MWh (B3).** Joint per-firm view of the  $q_2$  outcome (Ito–Reguant IDA upward adjustment) and DA cleared volume. The mechanism storyboard predicts *negative*  $\beta_3$  on DA cleared (firms withhold in DA in critical hours post-reform) and *positive*  $\beta_3$  on  $q_2$  (firms add upward in IDA). EDP-PT has no `pibci` entries (Portuguese-zone units do not appear in OMIE intraday programmes); the DA-side effect is identified.

| Firm   | $q_2$ outcome (B1) |        | DA cleared (B3) |         |
|--------|--------------------|--------|-----------------|---------|
|        | $\beta_3$          | SE     | $\beta_3$       | SE      |
| IB     | +13.66***          | (3.68) | −40.09***       | (11.70) |
| GE     | +14.56***          | (2.68) | +6.96           | (3.70)  |
| GN     | +0.63              | (1.78) | −28.47***       | (7.10)  |
| HC     | +6.64***           | (1.86) | −3.19           | (2.54)  |
| EDP-PT | —                  | —      | −118.51***      | (7.85)  |

## 5.2 DA → IDA price wedge: supporting outcome

**Table 5: B2 – Within-day DiD on the DA → IDA price wedge.** Outcome is hourly Spain DA clearing price minus IDA clearing price (averaged across IDA sessions) in EUR/MWh. Same-calendar-month restriction (Oct-Dec 2024 vs Oct-Dec 2025) shows the within-day differential expanded modestly post-reform. The full-window specification (2024-01 to 2025-12) attributes most of the within-day wedge spike to the post-blackout *operación reforzada* (April–September 2025): once the reforzada × critical interaction is included, the post-MTU15-DA × critical coefficient becomes large and negative, indicating that MTU15-DA itself *reduces* the within-day wedge after controlling for the regulatory overlay. The DA-IDA wedge is therefore a noisy outcome for MTU15-DA identification absent reforzada controls.

| Specification                               | $\beta_3$ | SE     | $p$    | $G$ |
|---------------------------------------------|-----------|--------|--------|-----|
| Same-cal-month (Oct-Dec 2024 vs 2025)       | +4.62**   | (1.43) | 0.0012 | 184 |
| Full window, no controls                    | +0.79     | (1.18) | 0.5043 | 730 |
| Full window, + reforzada × crit             | −10.06*** | (1.39) | 0.0000 | 730 |
| Full window, + reforzada + MTU15-IDA × crit | −10.06*** | (1.39) | 0.0000 | 730 |

## 5.3 Other outcomes (DA cleared volume, bid-shape metrics)

## 5.4 Conditional parallel trends with renewable and reforzada controls

**Table 6: B4 – Conditional parallel trends: spec stack for  $q_2$ , treatment group only.**  $\beta_3$  stability under successive control additions, starting from the baseline B1 specification (firm + DOW fixed effects). Spec (2) adds Spain hourly wind and solar production levels (ENTSO-E A75); Spec (3) adds the interaction of critical-hour dummy with VRE production (allowing the VRE effect to differ across critical and flat hours); Spec (4) adds calendar-month fixed effects. The same-calendar-month design already absorbs reforzada and MTU15-IDA via the post dummy (both hold constant within each window), so no separate controls for those reforms are needed in this spec stack. Stability of  $\beta_3$  across columns is the conditional parallel-trends defence of the headline result.

| Specification                | $\beta_3$ | SE     | $p$    | $G$ |
|------------------------------|-----------|--------|--------|-----|
| (1) Baseline (firm + DOW FE) | +7.56***  | (1.42) | 0.0000 | 184 |
| (2) + wind, solar levels     | +7.61***  | (1.45) | 0.0000 | 184 |
| (3) + crit × VRE             | +7.40***  | (1.52) | 0.0000 | 184 |
| (4) + calendar-month FE      | +7.40***  | (1.52) | 0.0000 | 184 |

## 6 Conclusion

### A Robustness checks

#### A.1 Sensitivity to critical-hours definition

#### A.2 Sensitivity to firm-partition (pivotality vs administrative)

#### A.3 Same-cal-month vs full-window comparisons

**Table 7: B5 – Robustness panel:  $\beta_3$  across specifications, treatment group on  $q_2$ .** Each row reports the headline coefficient  $\beta_3$  under a single sensitivity. Section B5.1 varies the critical-hours definition between four candidates (price\_peak is canonical). Section B5.2 varies the firm partition between the pivotality-based set (which includes EDP-PT) and the administrative IB/GE/GN/HC tier; the two coincide because EDP-PT lacks `pibci` data so its inclusion does not alter the `q_2` regression. Section B5.3/B5.5 varies the window: the full 2024–2025 panel reduces the coefficient because the pre-MTU15-DA half of the panel is post-MTU15-IDA, where  $q_2$  magnitudes are already crushed by the IDA reform; excluding the reforzada-active months (April–September 2025) recovers a coefficient close to the same-calendar-month estimate. Section B5.4 verifies that excluding individual units (EDP-PT, ABO2G) does not change the result.

| Sensitivity                             | $\beta_3$ | SE     | $p$    | $G$ |
|-----------------------------------------|-----------|--------|--------|-----|
| <i>B5.1 — Critical-hours definition</i> |           |        |        |     |
| price_peak h{18-22} (canonical)         | +8.78***  | (1.65) | 0.0000 | 184 |
| supply_ramp h{7,8,16,17,18}             | +6.77***  | (1.33) | 0.0000 | 184 |
| demand_peak h{16-20}                    | +8.43***  | (1.49) | 0.0000 | 184 |
| joint h{7-8, 16-22}                     | +7.56***  | (1.42) | 0.0000 | 184 |
| <i>B5.2 — Firm partition</i>            |           |        |        |     |
| Pivotality-based treatment set          | +7.56***  | (1.42) | 0.0000 | 184 |
| Administrative IB/GE/GN/HC              | +7.56***  | (1.42) | 0.0000 | 184 |
| <i>B5.3 / B5.5 — Window</i>             |           |        |        |     |
| Full window 2024-2025                   | +3.40***  | (0.41) | 0.0000 | 730 |
| Full window minus Apr-Sep 2025          | +6.61***  | (0.71) | 0.0000 | 575 |
| <i>B5.4 — Sample exclusions</i>         |           |        |        |     |
| Drop EDP-PT                             | +7.56***  | (1.42) | 0.0000 | 184 |
| Drop ABO2G                              | +7.56***  | (1.42) | 0.0000 | 184 |

#### A.4 Placebo specifications

#### A.5 Operational vs strategic decomposition (PIBCA vs PHF)

#### A.6 Bid-shape robustness in the competitive-zone restriction

### B Data appendix

#### B.1 Sequential-market technical reference (PDBC, PDBF, PDVD, PIBCA, PHF)

#### B.2 File-format specifications and parser conventions

#### B.3 Power-vs-energy unit discipline

#### B.4 Coverage and missingness

### References

- Blaise Allaz and Jean-Luc Vila. Cournot competition, forward markets and efficiency. *Journal of Economic Theory*, 59(1):1–16, 1993.
- James B. Bushnell. A mixed complementarity model of hydro-thermal electricity competition in the western US. *Operations Research*, 51(1):80–93, 2003.
- Clément de Chaisemartin and Xavier D’Haultfoe uille. Two-way fixed effects estimators with heterogeneous treatment effects. *American Economic Review*, 110(9):2964–96, 2020.
- Ali Hortaçsu and Steven L. Puller. Understanding strategic bidding in multi-unit auctions: a case study of the texas electricity spot market. *RAND Journal of Economics*, 39(1):86–114, 2008.
- Koichiro Ito and Mar Reguant. Sequential markets, market power, and arbitrage. *American Economic Review*, 106(7):1921–57, 2016.
- Erin T. Mansur. Measuring welfare in restructured electricity markets. *Review of Economics and Statistics*, 90(2):369–386, 2008.
- Frank A. Wolak. Measuring unilateral market power in wholesale electricity markets: the california market, 1998–2000. *American Economic Review*, 93(2):425–430, 2003.